

COMP 1110	
Term	Fall 2024
Course Name	Introduction to Computer Programming
Assignment	2
Due Date	October 25, 2024 @23:59
Marks	30 (zero mark for late submission)
Submission Instructions	1. Your program should be well-written 2. Submit Jupiter notebook file or .py file and name it as FirstNameLastNameAssignment-2 <i>Additional instructions:</i> 1. Add proper comments to your code to enhance readability and explain what the code is doing 2. Use the Camel case naming convention for naming the variable names.

Note: When writing your Python code, please remember to use meaningful variable names that clearly describe their purpose. Additionally, include proper comments throughout your code to explain complex logic or describe the functionality of specific sections. Furthermore, for each function, write a docstring at the beginning.

Q1: (5 points)

Write a function that asks the user to enter a distance in kilometers, then convert that distance to miles. The conversion can be done by using the formula:

Miles = Kilometers * 0.6214

Expected Output:

```
Enter distance in kilometers: 10
10.0 kilometers is equal to 6.214 miles.
```

Q2: (7 points)

Write a program (**with functions**) that gives simple math quizzes. The program should display two random numbers that are to be added, such as:

```
279
+ 190
```

The program should allow the student to enter the answer. If the answer is correct, a message of congratulations should be displayed. If the answer is incorrect, a message showing the correct answer should be displayed.

Expected Output:

```
298
+ 66
Enter your answer: 364
Congratulations! Your answer is correct.
```

```
499
+ 53
Enter your answer: 520
Sorry, the correct answer is 552.
```

Q3: (6 points)

Write a function called *who_am_i* that asks the user 3 yes/no questions to guess their personality type. Based on the answers, print a personality trait.

Questions:

- Do you like being around a lot of people? (yes/no)

- Do you enjoy trying new things? (yes/no)
- Do you prefer to plan things out? (yes/no)

Based on the answers:

- If they answer "yes" to all, print "You are adventurous and outgoing!"
- If they answer "no" to all, print "You are thoughtful and reserved."
- If they have a mix of answers, print "You have a balanced personality."

Expected Output:

```
Do you like being around a lot of people? (yes/no): yes
Do you enjoy trying new things? (yes/no): yes
Do you prefer to plan things out? (yes/no): yes
You are adventurous and outgoing!
```

Q4: (5 points)

Write a function called *fizz_buzz* that prints numbers from 1 to 20. For multiples of 3, print "Fizz" instead of the number, and for multiples of 5, print "Buzz." For numbers that are multiples of both 3 and 5, print "FizzBuzz."

Expected output:

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
```

Q5: (7 points)

Write a function called `find_min_max`. This function should ask the user to enter numbers one by one. When the user enters 0, the input will stop. The function should find and show the smallest and largest numbers entered but ignore the 0.

Expected Output:

```
Enter a number (0 to end): 67
Enter a number (0 to end): 98
Enter a number (0 to end): 09
Enter a number (0 to end): 99
Enter a number (0 to end): 76
Enter a number (0 to end): 88
Enter a number (0 to end): 45
Enter a number (0 to end): 0
Minimum number: 9
Maximum number: 99
```